Connor Williams
Assignment 4

Question 1:

```sql
#Question 1: Retrieve the mission name, rocket name,
#organization name, launch site name, destination, launch year, and outcome
-- JOIN connects missions to their rockets, organizations and launch sites
-- Primary Keys: missions.missionid, rockets.rocketid, organizations.orgid and launch_sites.siteid
-- Foreign Keys: missions.rocketid, missions.siteid and rockets.orgid
-- ORDER BY sorts by newest launch year first, then mission name A-Z
SELECT missions.missionname,
       rockets.rocketname,
       organizations.orgname,
       launch_sites.sitename,
       missions.destination,
       missions.launchyear,
       missions.outcome
FROM missions
JOIN rockets
    ON missions.rocketid = rockets.rocketid
JOIN organizations
    ON rockets.orgid = organizations.orgid
JOIN launch_sites
    ON missions.siteid = launch_sites.siteid
ORDER BY missions.launchyear DESC,
         missions.missionname ASC;
```

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Result Grid Filter Rows: Search Export:

missionname	rocketname	orgname	sitename	destination	launchyear	outcome
Artemis I	Saturn V	National Aeronautics and Space Administration	Kennedy Space Center	Moon	2024	Success
Chang'e 6	Long March	China National Space Administration	Jiuquan Satellite Launch Center	Moon	2024	Success
Crew-8	Falcon 9	Space Exploration Technologies Corp	Kennedy Space Center	ISS	2024	Success
Gaganyaan Test	PSLV	Indian Space Research Organisation	Satish Dhawan Space Centre	LEO	2024	Success
HTV-X1 Test	H-IIB	Japan Aerospace Exploration Agency	Tanegashima Space Center	ISS	2024	Success
Progress MS-27	Proton-M	Roscosmos State Corporation	Baikonur Cosmodrome	ISS	2024	Success
Sentinel-6B	Falcon 9	Space Exploration Technologies Corp	Vandenberg Space Force Base	LEO	2024	Success
StriX-1	Electron	Rocket Lab	Mahia Launch Complex 1	LEO	2024	Success

Question 2:

```sql
#Question 2:
#Retrieve the mission name, rocket name, payload name,
#payload type, and payload mass (in kg) for all payloads launched in 2024
-- Primary Keys: payloads.payloadid, missions.missionid, rockets.rocketid
-- Foreign Keys: payloads.missionid and missions.rocketid
-- JOIN connects payloads to their missions and rockets and WHERE only includes missions launched in 2024
-- ORDER BY sorts payloads from heaviest to lightest
SELECT missions.missionname,
       rockets.rocketname,
       payloads.payloadname,
       payloads.payloadtype,
       payloads.masskg
FROM payloads
JOIN missions
    ON payloads.missionid = missions.missionid
JOIN rockets
    ON missions.rocketid = rockets.rocketid
WHERE missions.launchyear = 2024
ORDER BY payloads.masskg DESC;

#Question 3:
#Retrieve all launch sites along with their matching mission names,
#rockets, and launch years for missions that took place in 2024
```

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Result Grid Filter Rows: Search Export:

missionname	rocketname	payloadname	payloadtype	masskg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450
Chang'e 6	Long March	Queqiao-2 Relay	Sattelite	1200
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Sattelite	1192
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Sattelite	100

Question 3:

```sql
#Question 3:
#Retrieve all launch sites along with their matching mission names,
#rockets, and launch years for missions that took place in 2024
-- Primary Keys: launch_sites.siteid, missions.missionid and rockets.rocketid
-- Foreign Keys: missions.siteid and missions.rocketid
-- LEFT JOIN includes launch sites even if they have no 2024 missions
-- Null values are shown
-- ORDER BY sorts launch sites alphabetically
SELECT launch_sites.sitename,
       missions.missionname,
       rockets.rocketname,
       missions.launchyear
FROM launch_sites
LEFT JOIN missions
    ON launch_sites.siteid = missions.siteid
    AND missions.launchyear = 2024
LEFT JOIN rockets
    ON missions.rocketid = rockets.rocketid
ORDER BY launch_sites.sitename ASC;


#Question 4:
```

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Result Grid Filter Rows: Search Export:

sitename	missionname	rocketname	launchyear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Kennedy Space Center	Artemis I	Saturn V	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

Result 11

Question 4:

```sql
-- SUM adds payload masses, and also GROUP BY groups them by organization
WITH OrganizationPayload AS (
    SELECT organizations.orgid,
           SUM(payloads.masskg) AS TotalPayloadMass
    FROM organizations
    JOIN rockets
        ON organizations.orgid = rockets.orgid
    JOIN missions
        ON rockets.rocketid = missions.rocketid
    JOIN payloads
        ON missions.missionid = payloads.missionid
    GROUP BY organizations.orgid
)

SELECT organizations.orgname,
       organizations.headquarterscountry,
       organizations.orgtype,
       OrganizationPayload.TotalPayloadMass
FROM organizations
JOIN OrganizationPayload
    ON organizations.orgid = OrganizationPayload.orgid
ORDER BY OrganizationPayload.TotalPayloadMass DESC;
```

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Result Grid Filter Rows: Search Export:

orgname	headquarterscoun...	orgtype	TotalPayloadMass
National Aeronautics and Space Administration	USA	Government	25800
Space Exploration Technologies Corp	USA	Private	13692
Japan Aerospace Exploration Agency	JPN	Government	6000
Roscosmos State Corporation	RUS	Government	2450
China National Space Administration	CHN	Government	1200
Indian Space Research Organisation	IND	Government	150
Rocket Lab	NZL	Private	100

Question 5:

```sql
WITH CountryMissions AS (
    SELECT launch_sites.countrycode,
           COUNT(missions.missionid) AS TotalMissions
    FROM launch_sites
    JOIN missions
        ON launch_sites.siteid = missions.siteid
    GROUP BY launch_sites.countrycode
),

CountryPayload AS (
    SELECT launch_sites.countrycode,
           SUM(payloads.masskg) AS TotalPayloadMass
    FROM launch_sites
    JOIN missions
        ON launch_sites.siteid = missions.siteid
    JOIN payloads
        ON missions.missionid = payloads.missionid
    GROUP BY launch_sites.countrycode
)

SELECT countries.countryname,
       COALESCE(CountryMissions.TotalMissions, 0) AS TotalMissions,
       COALESCE(CountryPayload.TotalPayloadMass, 0) AS TotalPayloadMass
FROM countries
LEFT JOIN CountryMissions
    ON countries.countrycode = CountryMissions.countrycode
```

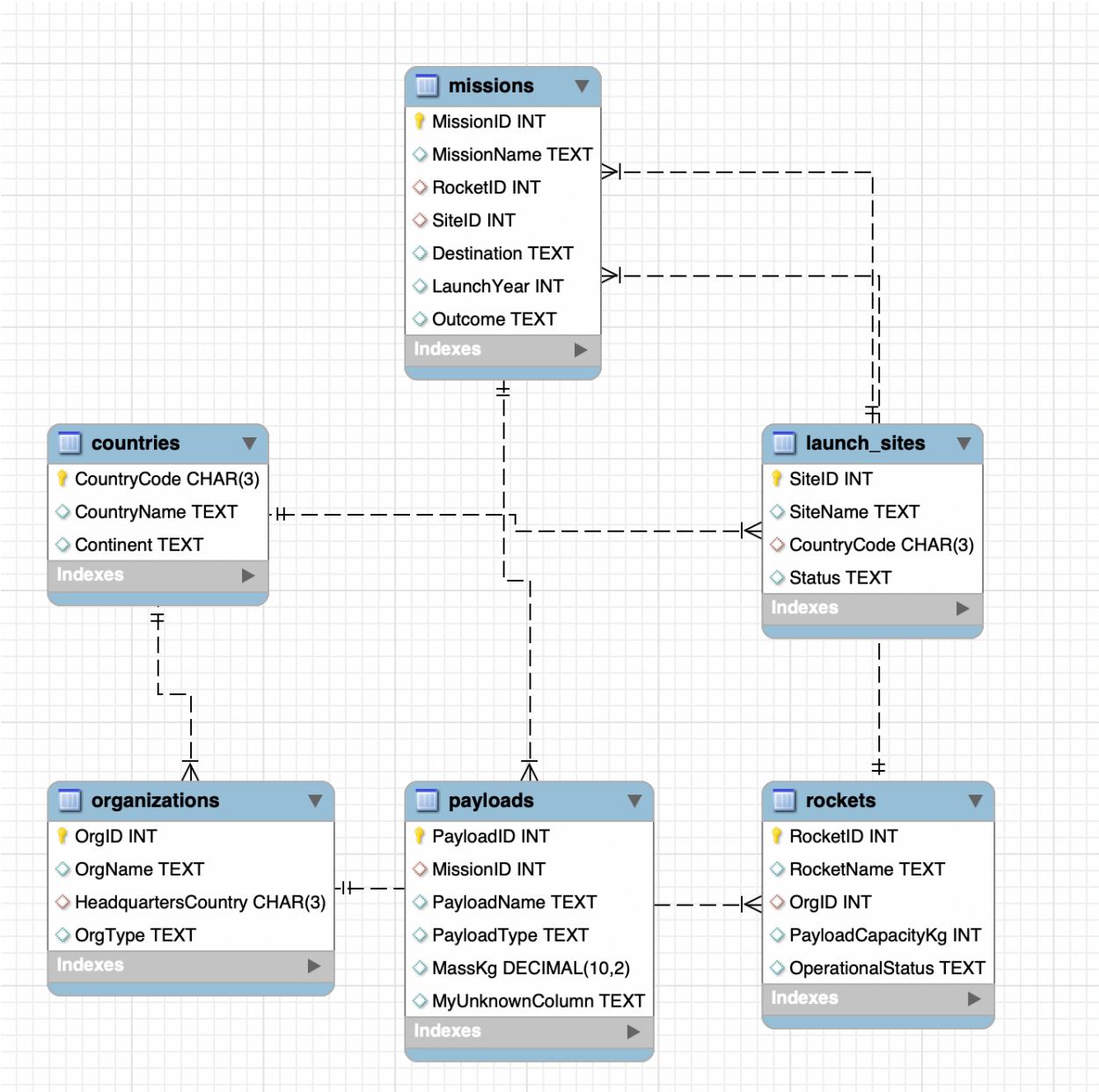
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Result Grid Filter Rows: Search Export:

countryname	TotalMissions	TotalPayloadMass
United States	3	39492
Japan	1	6000
Russia	1	2450
China	1	1200
India	1	150
New Zealand	1	100
France	0	0
United Kingdom	0	0

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Diagram:


missions
MissionID INT
MissionName TEXT
RocketID INT
SiteID INT
Destination TEXT
LaunchYear INT
Outcome TEXT
Indexes
countries
CountryCode CHAR(3)
CountryName TEXT
Continent TEXT
Indexes
launch_sites
SiteID INT
SiteName TEXT
CountryCode CHAR(3)
Status TEXT
Indexes
organizations
OrgID INT
OrgName TEXT
HeadquartersCountry CHAR(3)
OrgType TEXT
Indexes
payloads
PayloadID INT
MissionID INT
PayloadName TEXT
PayloadType TEXT
MassKg DECIMAL(10,2)
MyUnknownColumn TEXT
Indexes
rockets
RocketID INT
RocketName TEXT
OrgID INT
PayloadCapacityKg INT
OperationalStatus TEXT
Indexes